# UMAT documentation

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# Chapter 1

# Introduction

The commercial finite-element solver ABAQUS [1] offers interfaces that make possible to implement many mechanical constitutive models of arbitrary complexity depending on various materials. In case the available ABAQUS material library is insufficient to represent particular material behavior, user-defined material models, namely subroutine UMAT and subroutine VUMAT are incorporated in ABAQUS/Standard and ABAQUS/Explicit, respectively.

In this guide, an internally-developed user subroutine UMAT is introduced, which includes:

• local crystal plasticity models for basic face-centered cubic (FCC) and body-centered cubic (BCC) metals,

• non-local crystal plasticity models for the influence of heterogeneous plastic deformation caused by small scale deformation (e.g., nanoindentation) or micro-structural features (e.g., phase transformation in TRIP assisted steels) in metals,

• creep models for single crystal superalloys and nickel single crystals.

These constitutive models describe the material behavior on each Gaussian point. Through an implicit scheme method (Newton-Raphson algorithms), numerical approximations to the solution of the time-dependent constitutive equations are obtained [2, 3]. The detailed elaboration of this UMAT is shown in the following chapters.

# Chapter 2

# UMAT structure

## 2.1 UMAT files and subroutines

- umat.f
  This file establishes the FORTRAN environment to link the UMAT subroutine to ABAQUS and arranges running process of UMAT.
  - subroutine uexternaldb
  - subroutine umat
- mod_alloys.f
  In this file, all material parameters and constitutive equations are defined. The following materials are included: Aluminum, Copper, Ferrite, Austenite, Single Crystal Superalloys and Nickel.
  - subroutine sub_constant_ini_Aluminum
  - subroutine sub_get_parameter_Aluminum
  - subroutine sub_flow_harden_Aluminum
  - subroutine sub_constant_ini_Copper
  - subroutine sub_get_parameter_Copper
  - subroutine sub_flow_harden_Copper
  - subroutine sub_constant_ini_Ferrite
  - subroutine sub_get_parameter_Ferrite
  - subroutine sub_flow_harden_Ferrite
  - subroutine sub_constant_ini_Austenite

  - subroutine sub_get_parameter_Austenite
  - subroutine sub_flow_harden_Austenite
  - subroutine sub_constant_ini_Superalloy
  - subroutine sub_get_parameter_Superalloy
  - subroutine sub_flow_harden_Superalloy
  - subroutine sub_constant_ini_Nickel
  - subroutine sub_get_parameter_Nickel
  - subroutine sub_flow_harden_Nickel
  - subroutine mod_alls_ini
  - subroutine get_mat_parameter
  - subroutine cal_flow_harden

- mod_gaussp.f
  In this file, general variables related to continuum mechanics, microstructure, Newton-Raphson algorithms and initial constants are defined.

  - subroutine mod_gspt_ini

- mod_wkcoup.f
  This file incorporates the effects of distinct additional deformation mechanisms; such as the strain gradients induced by the evolution of the geometrically necessary dislocation (GND) density, the transformation-induced plasticity (TRIP) mechanism, the internal stresses due to lattice mismatch and deformation heterogeneity between distinct phases, and other softening and hardening effects in superalloys.

  - subroutine sub_grad_ini
  - subroutine sub_grad_evolution
  - subroutine sub_grad_effect
  - subroutine sub_trip_ini
  - subroutine sub_trip_evolution
  - subroutine sub_trip_effect
  - subroutine sub_int_ini
  - subroutine sub_int_evolution
  - subroutine sub_int_effect
  - subroutine sub_sup_ini

  - subroutine sub_sup_evolution
  - subroutine sub_sup_effect
  - subroutine mod_wkcp_ini
  - subroutine wkcoup_evolution
  - subroutine wkcoup_effect

- mod_stress.f
  In this file, two methods (explicit and implicit) are introduced to update stresses, deformation gradient and hardening variables.

  - subroutine cal_stress_add (explicit)
  - subroutine cal_stress_mul (implicit)
  - subroutine cal_NRmatrix
  - subroutine cal_MatStiffness

- mod_smthgd.f
  In this file, the Fast Fourier Transformation (FFT) method is implemented to calculate the first and second order of the strain gradient.

  - subroutine cal_Fp_grad
  - subroutine cal_V_smth
  - subroutine cal_V_grad

- other_code.f
  This file offers a number of auxiliary subroutines to support some additional mathematical functions.

  - subroutine caleulang
  - subroutine icams_misori
  - subroutine icams_determ
  - subroutine icams_Eang2Q
  - subroutine icams_Q2Eang
  - subroutine pm
  - subroutine gaussj
  - subroutine fourn
  - subroutine icams_conv33to6

  - subroutine icams_conv33to9
  - subroutine icams_conv6to33
  - subroutine stress_dislocation

- Matrix.dat
  This file is used to calculate internal stresses in single crystal superalloys.

## 2.2 Basic procedure

The UMAT performs transformations of general constitutive rate equations into incremental equations through an implicit scheme method (Newton-Raphson) to obtain numerical approximations to the solution of the time-dependent crystal plasticity equations [2]. The UMAT subroutine is called locally on each integration point (Gaussian/material point) to calculate stresses, strains and all solution-dependent state variables. The variable "lop" in the subroutine **uexternaldb** controlls the analysis procedure. As shown in Figure 2.1, after the initialization, the material deformation analysis begins with the start of increment. Through the Newton-Raphson iteration, the reasonable solutions of incremental equations are obtained. At the end of each increment, all state variables are updated.

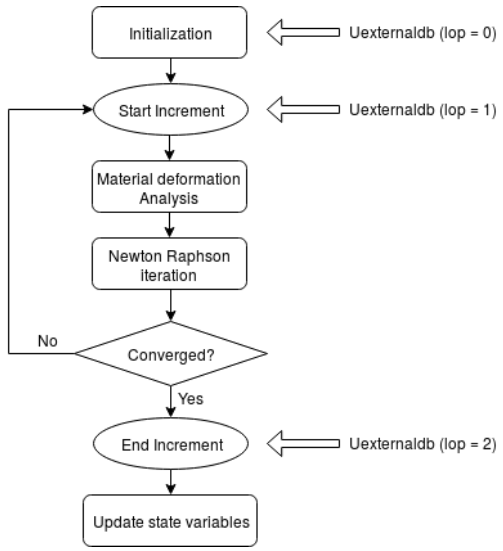


Figure 2.1: Simplified flow chart of ABAQUS/Standard analysis with UMAT.

# Chapter 3

# UMAT description

## 3.1 Initialization

### 3.1.1 Constants and material parameters

All the constants and continuum mechanics variables associated to one Gaussian point are defined in the module **mod_gaussp** in **mod_gaussp.f**.

The material parameters are initialized at the beginning of each material module in **mod_alloys.f**. The slip systems and Schmid matrix are defined in the general subroutine **mod_alls_ini**, which includes:

- sub_constant_ini_Aluminum
- sub_constant_ini_Copper
- sub_constant_ini_Ferrite
- sub_constant_ini_Austenite
- sub_constant_ini_Superalloy
- sub_constant_ini_Nickel

Based on the microstructures of different materials (fcc, bcc), the slip direction $\mathbf{d}$ and slip plane normal $\mathbf{n}$ are defined. Thus, the Schmid matrix is given by

$$\mathbf{M} = \mathbf{d} \otimes \mathbf{n}. \tag{3.1}$$

The symmetric part of the Schmid matrix $\mathbf{M}_\mathrm{s}$ and anti-symmetric part of the Schmid matrix $\mathbf{M}_\mathrm{a}$ are derived as

$$\mathbf{M}_\mathrm{s} = \frac{1}{2}(\mathbf{M} + \mathbf{M}^\mathrm{T}), \tag{3.2}$$

$$\mathbf{M}_{\mathrm{a}} = \frac{1}{2}(\mathbf{M} - \mathbf{M}^{\mathrm{T}}). \tag{3.3}$$

According to the ABAQUS stress sequence vectors (IB1=[1,2,3,1,1,2,2,3,3] and IB2=[1,2,3,2,3,3,1,1,2]) defined in the subroutine **mod_gspt_ini** in **mod_gaussp.f**, the Schmid matrix, its symmetric and anti-symmetric expressions are transformed into a vectorial notation considering the diagonal components of the Schmid matrix $\mathbf{M}_{\mathrm{s}}$ and two times the off-diagonal components of the Schmid matrix $\mathbf{M}_{\mathrm{s}}$, respectively.

All materials use the simplest anisotropic case of cubic symmetry in which three independent elastic constants (c11, c12 and c44) are prescribed to define the stiffness matrix $\mathbf{K}$:

$$\mathbf{K} = \begin{pmatrix} c11 & c12 & c12 & 0 & 0 & 0 \\ c12 & c11 & c12 & 0 & 0 & 0 \\ c12 & c12 & c11 & 0 & 0 & 0 \\ 0 & 0 & 0 & c44 & 0 & 0 \\ 0 & 0 & 0 & 0 & c44 & 0 \\ 0 & 0 & 0 & 0 & 0 & c44 \end{pmatrix}.$$

The contribution of a certain slip system $\beta$ on the hardening of a slip system $\alpha$ is related to the cross hardening matrix $q_{\alpha\beta}$, which is defined as a latent-hardening matrix Hmij in the UMAT. The value of components in Hmij are 1.0 for coplanar slip systems or 1.4 for non-coplanar systems.

Through the subroutine **get_mat_parameter** in **mod_alloys.f**, the initialized material parameters in the subroutine **sub_constant_ini_"material"** are assigned to all the slip systems. For different materials, the subroutine **get_mat_parameter** contains:

- sub_get_parameter_Aluminium
- sub_get_parameter_Copper
- sub_get_parameter_Ferrite
- sub_get_parameter_Austenite
- sub_get_parameter_Superalloy
- sub_get_parameter_Nickel

### 3.1.2 Weak coupling parameters

The state variables associated to distinct additional deformation mechanisms are initialized in the subroutine **mod_wkcp_ini** in **mod_wkcoup.f**. The corresponding material parameters to different deformation mechanisms are set in the beginning of each weak coupling module.

## 3.2 Selection of material module and weak coupling module

There are 6 material modules corresponding to Aluminum (ialloy=1), Copper (ialloy=2), Ferrite (ialloy=3), Austenite (ialloy=4), Single Crystal Superalloys (ialloy=5) and Nickel (ialloy=6), which can be chosen by the parameter "ialloy". The distinct deformation mechanisms in **mod_wkcoup.f** can be combined with different materials.
If Iwkcoup_grad=1, the effect of strain gradients due to super GND is activated. (ialloy=1–4)
If Iwkcoup_trip=1, the effect of phase transformation (TRIP) is activated. (ialloy=4)
If Iwkcoup_int=1, the effect of internal stresses in superalloys is activated. (ialloy=5)
If Iwkcoup_sup=1, the additional softening and hardening effects in superalloys are activated. (ialloy=5)

## 3.3 Constitutive equations and hardening laws

The constitutive equations and hardening laws are defined in the subroutine **cal_flow_harden** in **mod_alloys.f**.

If ialloy=1–4, the common flow rule is

$$\dot{\gamma}^{\alpha} = \dot{\gamma}_0 \mid \frac{\tau^{\alpha}}{\tau^{\alpha}_{\text{eff}}} \mid^{p_1} \text{sign}(\tau^{\alpha}), \tag{3.4}$$

where $\dot{\gamma}^{\alpha}$ is the shear rate in a particular slip system $\alpha$, $\dot{\gamma}_0$ is the reference shear rate, $\tau^{\alpha}$ is the resolved shear stress, $\tau^{\alpha}_{\text{eff}}$ is the effective slip resistance, and $p_1$ is the inverse value of the strain rate sensitivity.

$$\tau^{\alpha}_{\text{eff}} = \hat{\tau}^{\alpha}_{\text{c}} + \tau^{\alpha}_{\text{wcp}}, \tag{3.5}$$

where $\tau^{\alpha}_{\text{wcp}}$ is the weak coupling stress which results from the strain gradient effect and TRIP effect. $\hat{\tau}^{\alpha}_{\text{c}}$ is determined by

$$\dot{\hat{\tau}}^{\alpha}_{\text{c}} = \sum_{\beta=1}^{N_{\text{slip}}} h_0 q_{\alpha\beta} \left(1 - \frac{\hat{\tau}^{\beta}_{\text{c}}}{\hat{\tau}_{\text{sat}}}\right)^{p_2} \left|\dot{\gamma}^{\beta}\right|, \tag{3.6}$$

where $N_{\text{slip}}$ is the number of slip systems, $h_0$ is the reference hardening parameter, $q_{\alpha\beta}$ is the cross hardening matrix, $\hat{\tau}_{\text{sat}}$ is the saturated slip resistance due to dislocation density accumulation, and $p_2$ is a fitting parameter.

If ialloy=5–6, the common flow rule for creep deformation is

$$\dot{\gamma}^{\alpha} = \dot{\gamma}_0 \cdot \exp\Big(-\frac{Q}{RT}\Big) \mid \frac{\tau^{\alpha}}{\tau^{\alpha}_{\text{eff}}} \mid^{p_1} \text{sign}(\tau^{\alpha}), \tag{3.7}$$

where $Q$ is the activation energy for dislocation slip, $R$ is the gas constant and $T$ is the temperature. In the $\gamma$ phase of superalloys,

$$\tau^{\alpha}_{\text{eff}} = \hat{\tau}^{\alpha}_{\text{c}} + \tau_{\text{oro}} - \hat{\tau}_{\text{soft}}. \tag{3.8}$$

$\tau_{\text{oro}}$ is the Orowan stress given by

$$\tau_{\text{oro}} = \sqrt{\frac{2}{3}}\frac{\mu b}{w}, \tag{3.9}$$

where $\mu$ is the shear modules, $b$ is the magnitude of Burgers vector and $w$ is the channel width. $\hat{\tau}_{\text{soft}}$ is the softening stress due to dislocation climb. In the $\gamma'$ phase of superalloys,

$$\tau^{\alpha}_{\text{eff}} = \hat{\tau}^{\alpha}_{\text{c}} - \hat{\tau}^{\alpha}_{\text{m}} + \hat{\tau}^{\alpha}_{\text{kw}}. \tag{3.10}$$

$\hat{\tau}^{\alpha}_{\text{m}}$ is related to the special reaction of matrix dislocations, which plays a reduction effect in precipitate hardening. $\hat{\tau}^{\alpha}_{\text{kw}}$ is the Kear-Wilsdorf lock effect.

## 3.4 Calculation of stresses

### 3.4.1 General aspects

Two decomposition methods are introduced in **mod_stress.f** to determine the stresses: the additive decomposition method (subroutine **cal_stress_add**) and the multiplicative decomposition method (subroutine **cal_stress_mul**).

For the additive decomposition, an explicit integration scheme is adopted, which is simple, but only conditionally stable. If the increment is not small enough, the solution would drift from the correct one. Therefore, it requires small time steps and could be time-consuming. An implicit integration scheme is usually employed for the multiplicative decomposition method. It allows to obtain a accurate solution with larger time steps. Both explicit and implicit schemes need the Newton-Raphson iteration to update the stresses in every time step. All of our present constitutive models are implemented in the implicit way.

In general, the deformation gradient $\mathbf{F}$ evolves by the velocity gradient $\mathbf{L}$ as

$$\dot{\mathbf{F}} = \mathbf{L}\mathbf{F}. \tag{3.11}$$

For the additive decomposition, the velocity gradient is decomposed into symmetric and anti-symmetric parts as

$$\mathbf{L} = \mathbf{D} + \mathbf{W}, \tag{3.12}$$

where the symmetric part $\mathbf{D}$ is known as the rate of deformation tensor and the anti-symmetric part $\mathbf{W}$ is known as the spin tensor. They can be expressed by

$$\mathbf{D} = (\mathbf{L} + \mathbf{L}^{\text{T}})/2, \tag{3.13}$$

$$\mathbf{W} = (\mathbf{L} - \mathbf{L}^{\text{T}})/2. \tag{3.14}$$

According to the present rotation matrix $\mathbf{Q}_{\text{m0}}$, the new Schmid matrix is given by

$$\mathbf{M}^{\alpha} = \mathbf{Q}_{\text{m0}} \cdot \mathbf{M}^{\alpha}_{0} \cdot \mathbf{Q}^{\text{T}}_{\text{m0}}, \tag{3.15}$$

thus, the resolved shear stress is

$$\tau^{\alpha} = \mathbf{T} \cdot \mathbf{M}^{\alpha}, \tag{3.16}$$

where $\mathbf{T}$ is the Cauchy stress. After determining the shear rate by the flow rule, the Cauchy stress and the slip resistance are updated by the Newton-Raphson method, and the plastic velocity gradient is obtained by

$$\mathbf{L}_{\mathrm{p}} = \sum_{\alpha=1}^{N_{\mathrm{slip}}} \dot{\gamma}^{\alpha} \mathbf{M}^{\alpha}. \tag{3.17}$$

The new rotation matrix is calculated by

$$\mathbf{Q}_{\mathrm{m}} = (\mathbf{I} + \mathbf{L} - \mathbf{L}_{\mathrm{p}} \cdot dt) \cdot \mathbf{Q}_{\mathrm{m0}}, \tag{3.18}$$

where $\mathbf{I}$ is the unit tensor and $dt$ is the time step. From $\mathbf{Q}_{\mathrm{m}}$ the new Euler angles and new elastic stiffness can be obtained.

In the multiplicative decomposition method, the deformation gradient $\mathbf{F}$ is decomposed into elastic and plastic parts as

$$\mathbf{F} = \mathbf{F}_{\mathrm{e}} \mathbf{F}_{\mathrm{p}}. \tag{3.19}$$

The second Piola-Kirchhoff stress $\mathbf{S}$ is

$$\mathbf{S} = \frac{\mathbb{C}}{2} (\mathbf{F}_{\mathrm{e}}^{\mathrm{T}} \mathbf{F}_{\mathrm{e}} - \mathbf{I}), \tag{3.20}$$

where $\mathbb{C}$ is the elastic stiffness, hence, the resolved shear stress is

$$\tau^{\alpha} = \mathbf{S} \cdot \mathbf{M}^{\alpha}. \tag{3.21}$$

After solving the second Piola-Kirchhoff stress and the slip resistance by the Newton-Raphson method, the plastic velocity gradient is calculated by Equation 3.17, and the new plastic gradient is

$$\mathbf{F}_{\mathrm{p}} = (\mathbf{I} + \mathbf{L}_{\mathrm{p}} \cdot dt) \cdot \mathbf{F}_{\mathrm{p0}}. \tag{3.22}$$

The Cauchy stress is

$$\mathbf{T} = \frac{1}{\det(\mathbf{F}_{\mathrm{e}})} \mathbf{F}_{\mathrm{e}} \cdot \mathbf{S} \cdot \mathbf{F}_{\mathrm{e}}^{\mathrm{T}} \tag{3.23}$$

Finally, the Jacobian matrix is calculated to updated the stiffness matrix. The detailed solution of each quantity is explained in [4].

### 3.4.2 Newton Raphson iteration

With the known values of the stress $\mathbf{S}^{t}$, the slip resistance for different slip systems $\hat{\boldsymbol{\tau}}^{t}$ and the deformation gradient $\mathbf{F}^{t}$ at a certain time step, the stress and the slip resistance in the next time step can be

$$\mathbf{S}^{t+dt} \approx \mathbf{S}_{\max}(\mathbf{F}^{t+dt}) - \dot{\mathbf{S}}(\mathbf{F}^{t+dt}, \mathbf{S}^{t+dt}, \hat{\boldsymbol{\tau}}^{t+dt}) dt, \tag{3.24}$$

$$\hat{\boldsymbol{\tau}}^{t+dt} = \hat{\boldsymbol{\tau}}^{t} + \dot{\hat{\boldsymbol{\tau}}}(\mathbf{S}^{t+dt}, \hat{\boldsymbol{\tau}}^{t+dt})dt. \tag{3.25}$$

In order to solve $\mathbf{S}^{t+dt}$ and $\hat{\boldsymbol{\tau}}^{t+dt}$, two residual vectors $\mathbf{G}_1$ and $\mathbf{G}_2$ are defined as

$$\mathbf{G}_1 = \mathbf{S}^{t+dt} - \mathbf{S}_{\max}(\mathbf{F}^{t+dt}) + \dot{\mathbf{S}}(\mathbf{F}^{t+dt}, \mathbf{S}^{t+dt}, \hat{\boldsymbol{\tau}}^{t+dt})dt, \tag{3.26}$$

$$\mathbf{G}_2 = \hat{\boldsymbol{\tau}}^{t+dt} - \hat{\boldsymbol{\tau}}^{t} - \dot{\hat{\boldsymbol{\tau}}}(\mathbf{S}^{t+dt}, \hat{\boldsymbol{\tau}}^{t+dt})dt. \tag{3.27}$$

By using the 1st order Taylor expansion, the increments of the stress and slip resistance can be

$$d\mathbf{S}^{t+dt} = -\Big(\frac{\partial \mathbf{G}_1}{\partial \mathbf{S}^{t+dt}}\Big)^{-1}\Big(\mathbf{G}_1 + \frac{\partial \mathbf{G}_1}{\partial \boldsymbol{\tau}^{t+dt}} d\boldsymbol{\tau}^{t+dt}\Big) \tag{3.28}$$

$$d\boldsymbol{\tau}^{t+dt} = -\Big(\frac{\partial \mathbf{G}_2}{\partial \boldsymbol{\tau}^{t+dt}}\Big)^{-1}\Big(\mathbf{G}_2 + \frac{\partial \mathbf{G}_2}{\partial \mathbf{S}^{t+dt}} d\mathbf{S}^{t+dt}\Big) \tag{3.29}$$

After solving these linear equations, the new approximate solutions are obtained as

$$\mathbf{S}^{t+dt} \Leftarrow \mathbf{S}^{t+dt} + d\mathbf{S}^{t+dt} \tag{3.30}$$

$$\boldsymbol{\tau}^{t+dt} \Leftarrow \boldsymbol{\tau}^{t+dt} + d\boldsymbol{\tau}^{t+dt}. \tag{3.31}$$

If the sum of the two residual vectors is smaller than a predefined Newton Raphson tolarance (i.e., $\|\mathbf{G}_1\| + \|\mathbf{G}_2\| \leqslant G_{\text{critical}}$), a convergence reaches. Thus, the stress and the slip resistance are updated for the next time step. The subroutine **cal_NRmatrix** in **mod_stress.f** is used to calculate the tangent matrix for Equations 3.28 and 3.29.

# Chapter 4

# Weak coupling additional deformation mechanisms

The file **mod_wkcoup.f** includes four additional deformation mechanisms:

- Isotropic and kinematic hardening caused by plastic strain gradients (based on geometrically necessary dislocation (GND) density) [5].
- Transformation-induced plasticity (TRIP) mechanism [6].
- Internal stresses due to lattice mismatch and deformation heterogeneity in single crystal superalloys [7].
- Effects of the dislocation climb, the dislocation ribbons formation and the Kear-Wilsdorf lock in single crystal superalloys.

In the first module **mod_mesh_grid_size** in the file **mod_wkcoup.f**, the following parameters are defined:

- Number of elements.
- Number of Gaussian points in each element.
- Fast Fourier transformation grids.
- With/without a certain deformation mechanism.

## 4.1 Strain gradients due to GND

This module **mod_wkcoup_grad** is used to implement the isotropic and kinematic hardening caused by plastic strain gradients based on the concept of super GND densities and super GND segment pile-ups.

### 4.1.1 Parameters and initialization

The mechanism-related parameters are defined at the beginning of module **mod_wkcoup_grad**. In the following subroutine **sub_grad_ini**, the variables of the initial plastic deformation gradient, deformation gradient rate, first and second order deformation gradients are initiated for FEM Gaussian points and FFT grids.

### 4.1.2 Strain gradient evolution

In order to calculate the strain gradient, the FFT method is used in this model. In the subroutine **sub_grad_evolution**, a connection between FEM mesh and FFT periodic grid is established firstly. Meanwhile, the frequency space and the coordinates of FFT grids are computed. The first and second order gradient of the smoothed plastic deformation gradients are calculated by the subroutine **cal_Fp_grad** in **mod_smthgd.f**. The subroutine **cal_Fp_grad** starts by mapping the relevant variables from the FEM Gaussian points to FFT grids. Subsequently, the subroutine **cal_V_smth** is used to calculate a smoothed field of the strain gradients by solving a diffusion problem based on FFT method. Afterwards, the first and second order gradient of the smoothed field are computed by the subroutine **cal_V_grad**. Finally, these values of strain gradients on FFT grids are transformed to the FEM material points.

### 4.1.3 Hardenning effects

The subroutine **sub_grad_effect** in **mod_wkcoup.f** calculates the additional hardening effects due to the strain gradients. One is the isotropic hardening caused by super GNDs. Another is the kinematic hardening caused by super GND pile-ups.

As the model introduced by Ma et al. [5], the passing stress of GND densities is given by

$$\hat{\tau}^{\alpha}_{\mathrm{GNDi}} = c_1 \mu b \sqrt{\sum_{\beta=1}^{9} \chi_{\alpha\beta} \mid \bar{\rho}^{\beta} \mid} \tag{4.1}$$

where $c_1$ is the Taylor hardening coefficient, $\bar{\rho}$ is the GND density, and $\chi_{\alpha\beta}$ is the cross-hardening matrix between crystallographic mobile dislocations and super GNDs. This stress is represented by "IVB_gnd" in the code, which supplies the isotropic hardening.

In the presence of the second order gradient of $\mathbf{F}_{\mathrm{p}}$, the GNDs are heterogeneously distributed in the material. By assuming a cell with size $L^3$ at each material point $\kappa$, the super GND density

gradient is determined by the GND tensor with nine independent parts. It can be replaced by a dislocation segment producing the same dislocation density gradient at the center of the cell. The net Burgers vector of the dislocation segment is

$$\Delta\bar{\mathbf{b}}^{\alpha,I} = \frac{1}{2}(\bar{\rho}^{\alpha,I}\ L^3)\bar{\mathbf{d}}^{\alpha}, \tag{4.2}$$

where $\bar{\mathbf{d}}$ is the close-packed direction. The subroutine **stress_dislocation** in Module **other_code.f** is used to calculate the stress fields of a dislocation segment in an isotropic elastic medium at $\kappa$, based on the start position $\xi^{\alpha,I}$ and end position $\zeta^{\alpha,I}$ of the segment. Thus, a module $\mathbf{C}(\kappa,\ \xi^{\alpha,I},\ \zeta^{\alpha,I})_{ijk}$ (see Appendix [5]) is obtained, from which the internal stress in the reference configuration is formulated by

$$\mathbf{S}_{\mathrm{GND}} = \sum_{\alpha=1,I=1}^{9,3} \mathbf{C}(\kappa, \xi^{\alpha,I}, \zeta^{\alpha,I})\Delta\bar{\mathbf{b}}^{\alpha,I}, \tag{4.3}$$

and the internal stress in the intermediate configuration is

$$\widetilde{\mathbf{S}}_{\mathrm{GND}} = \mathbf{F}_{\mathrm{p}}\ \mathbf{S}_{\mathrm{GND}}\ \mathbf{F}_{\mathrm{p}}^{\mathrm{T}}. \tag{4.4}$$

The back stress $\tau^{\alpha}_{\mathrm{GNDk}}$ can be calculated by

$$\tau^{\alpha}_{\mathrm{GNDk}} = \widetilde{\mathbf{S}}_{\mathrm{GND}} \cdot \mathbf{M}^{\alpha}, \tag{4.5}$$

which is the kinematic hardening represented by "pk2i_gnd" in the code. Adding the isotropic and kinematic hardening into the flow rule, the shear rate is

$$\dot{\gamma}^{\alpha} = \dot{\gamma}_0 \left| \frac{\tau^{\alpha} + \tau^{\alpha}_{\mathrm{GNDk}}}{\hat{\tau}^{\alpha} + \hat{\tau}^{\alpha}_{\mathrm{GNDi}}} \right|^{p1} \mathrm{sign}(\tau^{\alpha} + \tau^{\alpha}_{\mathrm{GNDk}}). \tag{4.6}$$

## 4.2 Transformation-induced plasticity (TRIP)

The transformation induced plasticity (TRIP) effect is promoted for the phase transformation from retained Austenite (fcc) into Martensite (bct) under certain boundary conditions. As proposed by Ma et al. [6], the model adopts the inverse Nishiyama–Wassermann (NW) orientation relationship, sets explicit connections between the strain-induced martensite nucleation and the plastic deformation of austenite, as well as considers strain-induced martensite nucleation and stress-assisted martensite growth at the same time.

### 4.2.1 Parameters and initialization

According to Ma et al. [6], the Bain deformation matrix, the rotation matrix, phase transformation systems, the matrix for projection of strain from fcc slip systems to fault-band systems and the matrix for fault-band intersection mechanism are defined in the subroutine **sub_trip_ini**.

### 4.2.2 Eigendeformation and hardening due to TRIP

In the subroutine **sub_trip_effect**, all material parameters related to the TRIP mechanisms are listed firstly. Combining the martensite transformation, the total deformation gradient of the retained austenite is separated into an elastic deformation part $\mathbf{F}_{\mathrm{e}}$, a transformation part $\mathbf{F}_{\mathrm{tr}}$ and a plastic deformation part $\mathbf{F}_{\mathrm{p}}$ as

$$\mathbf{F} = \mathbf{F}_{\mathrm{e}}\ \mathbf{F}_{\mathrm{tr}}\ \mathbf{F}_{\mathrm{p}}. \tag{4.7}$$

The total martensite volume fraction includes the initial part, the strain-induced part and the stress-assisted part:

$$\boldsymbol{\eta} = \boldsymbol{\eta}^{0} + \boldsymbol{\eta}^{\mathrm{strain}} + \boldsymbol{\eta}^{\mathrm{stress}}. \tag{4.8}$$

The chemical driving force of the martensite transformation is determined by the chemical compositions and the activation temperature. Since the fault band intersections act as the preferential martensite nucleation sites, the probability rate of formating martensite nuclei in a transformation system $I$ by different fault bands is defined as

$$\dot{P}_I = H''_{I\alpha}\dot{\gamma}'_{\alpha} = H''_{I\beta}H'_{\beta\alpha}\dot{\gamma}_{\alpha}, \tag{4.9}$$

where $H''_{I\alpha}$ is the matrix for fault band intersection mechanism, $H'_{I\beta\alpha}$ is the matrix for projection of strain from fcc slip systems to fault-band systems, and $\dot{\gamma}_{\alpha}$ is the shear rate in the fcc slip system. The strain-induced martensite nucleation rate is given by

$$\dot{\eta}_I^{\mathrm{strain}} = c_{\mathrm{nucl}}\Big(1 - \sum_{J=1}^{N_{\mathrm{T}}} \eta_J\Big)\dot{P}_I\zeta_I^{\mathrm{strain}}, \tag{4.10}$$

where $c_{\mathrm{nucl}}$ justify the martensite nucleation kinetics, $N_{\mathrm{T}}$ is the total number of transformation systems, and $\zeta_I^{\mathrm{strain}}$ is the thermodynamic driving force determined by

$$\zeta_I^{\mathrm{strain}} = -[c_{\mathrm{dwp}}(2\eta_I - 6\eta_I^2 + 4\eta_I^3)] - [12\Delta G(\eta_I^2 - \eta_I^3)], \tag{4.11}$$

where $c_{\mathrm{dwp}}$ is the double well potential parameter and $\Delta G$ is the temperature-dependent bulk energy difference between martensite and austenite. The stress-assisted martensite growth rate is formulated as

$$\dot{\eta}_I^{\mathrm{stress}} = c_{\mathrm{pbmb}}(\eta_I^0 + \eta_I^{\mathrm{strain}})\Big(1 - \sum_{\beta=1}^{N_{\mathrm{T}}} \eta_\beta\Big)\zeta_I^{\mathrm{stress}}, \tag{4.12}$$

where $c_{\mathrm{pbmb}}$ is a temperature-dependent parameter controlling the martensite growth rate, and $\zeta_I^{\mathrm{stress}}$ is given by

$$\zeta_I^{\mathrm{stress}} = \det(\mathbf{F})\frac{\boldsymbol{\sigma}}{2}\Big(\mathbf{F}_{\mathrm{e}}\widetilde{\mathbf{N}}_I\mathbf{F}_{\mathrm{e}}^{-1} + \mathbf{F}_{\mathrm{e}}^{-\mathrm{T}}\widetilde{\mathbf{N}}_I^{\mathrm{T}}\mathbf{F}_{\mathrm{e}}^{\mathrm{T}}\Big) - [c_{\mathrm{dwp}}(2\eta_I - 6\eta_I^2 + 4\eta_I^3) + 12\Delta G(\eta_I^2 - \eta_I^3)], \tag{4.13}$$

where $\widetilde{\mathbf{N}}_I$ is the transformation system base tensor and $\boldsymbol{\sigma}$ is the Cauchy stress. The transformation part of deformation gradient $\mathbf{F}_{\mathrm{tr}}$ is

$$\mathbf{F}_{\mathrm{tr}} = \Big(\mathbf{I} + \sum_{I=1}^{N_{\mathrm{T}}} \Delta\eta_I^{\mathrm{stress}}\ \widetilde{\mathbf{N}}_I\Big)\ \mathbf{F}_{\mathrm{tr}}^{0}, \tag{4.14}$$

and the hardening rate due to martensite lamellar is

$$\dot{\hat{\tau}}^{\alpha}_{\mathrm{tr}} = c_{\mathrm{hard}}\mu \sum_{I=1}^{N_{\mathrm{T}}} \dot{\eta}_I \tag{4.15}$$

where $c_{\mathrm{hard}}$ is the lamellar hardening parameter. This hardening is represented by "IVB_trp" in the code.

## 4.3 Internal stresses due to lattice mismatch and deformation heterogeneity

When atoms are forced to be displaced from their equilibrium lattice positions, internal reaction forces are created giving rise to internal stresses [4]. In single crystal superalloys, the atoms of the two phases $\gamma$ and $\gamma'$ share one common lattice. However, between the two phases, due to slightly differences in the lattice parameters (lattice mismatch), coherent strains associated with high internal stresses are induced. The magnitude and sign of the lattice mismatch (Equation 4.16) determine the microstructure and its influence on dislocation interactions with $\gamma'$ precipitates. Therefore, the strengthening effect of $\gamma'$ precipitates has a strong dependence on the lattice mismatch $\delta$ [7].

$$\delta = 2\frac{a^{\gamma} - a^{\gamma'}}{a^{\gamma} + a^{\gamma'}} \tag{4.16}$$

where $a$ is the lattice parameter of $\gamma$ and $\gamma'$, respectively.

### 4.3.1 Parameters and initialization

In order to initialize the constants and state variables related to the additional eigenstrains produced due to the internal stresses in single crystal superalloys, the subroutine **sub_int_ini** is called. First, the subroutine defines the initial eigenstrain $\boldsymbol{\epsilon}$ as a six-component vector with the values of lattice mismatch between $\gamma$ and $\gamma'$ phases in the material points of $\gamma'$ phase. As the initial lattice mismatch usually varies from -0.005 to 0.005 for the case of single crystal Ni-base superalloys [8], the subroutine assigns the value of -0.005 to the first three components of vector accounting the $\gamma$ phase and the value of 0 to the rest of the components associated with the $\gamma'$ phase and cube slip systems: $\boldsymbol{\epsilon} = \big($ -0.0015, -0.0015, -0.0015, 0., 0., 0. $\big)$.

A linear relation between the internal stresses $\mathbf{S}^{\mathrm{int}}$ and eigenstrains is considered based on the Eshelby inclusion approach [9].

$$\mathbf{S}^{\mathrm{intx}} = \mathbb{C}''_{\mathrm{xx}}\boldsymbol{\epsilon}^{\mathrm{x}} + \mathbb{C}''_{\mathrm{xy}}\boldsymbol{\epsilon}^{\mathrm{y}} + \mathbb{C}''_{\mathrm{xz}}\boldsymbol{\epsilon}^{\mathrm{z}} + \mathbb{C}''_{\mathrm{xp}}\boldsymbol{\epsilon}^{\mathrm{p}} \tag{4.17}$$

$$\mathbf{S}^{\mathrm{inty}} = \mathbb{C}''_{\mathrm{yx}}\boldsymbol{\epsilon}^{\mathrm{x}} + \mathbb{C}''_{\mathrm{yy}}\boldsymbol{\epsilon}^{\mathrm{y}} + \mathbb{C}''_{\mathrm{yz}}\boldsymbol{\epsilon}^{\mathrm{z}} + \mathbb{C}''_{\mathrm{yp}}\boldsymbol{\epsilon}^{\mathrm{p}} \tag{4.18}$$

$$\mathbf{S}^{\mathrm{intz}} = \mathbb{C}''_{\mathrm{zx}}\boldsymbol{\epsilon}^{\mathrm{x}} + \mathbb{C}''_{\mathrm{zy}}\boldsymbol{\epsilon}^{\mathrm{y}} + \mathbb{C}''_{\mathrm{zz}}\boldsymbol{\epsilon}^{\mathrm{z}} + \mathbb{C}''_{\mathrm{zp}}\boldsymbol{\epsilon}^{\mathrm{p}} \tag{4.19}$$

$$\mathbf{S}^{\mathrm{intp}} = \mathbb{C}''_{\mathrm{px}}\boldsymbol{\epsilon}^{\mathrm{x}} + \mathbb{C}''_{\mathrm{py}}\boldsymbol{\epsilon}^{\mathrm{y}} + \mathbb{C}''_{\mathrm{pz}}\boldsymbol{\epsilon}^{\mathrm{z}} + \mathbb{C}''_{\mathrm{pp}}\boldsymbol{\epsilon}^{\mathrm{p}} \tag{4.20}$$

where the 16 components of $\mathbb{C}$ imported in code as MatrixA have a dependence on the unit cell geometry and elastic constants of the $\gamma$-phase and $\gamma'$-precipitate [10]. A FFT method [11, 7] together with a perturbation method were used to generate the MartixA. The construction of this matrix was based on the assumption of periodic boundary conditions, as the RVE was treated as an inclusion, and the rafting effect of $\gamma'$-precipitate was neglected [9]. Finally, the plastic deformation gradient $\mathbf{F}_{\mathrm{p}}$ and the volume fractions for each subsection of RVE are initialized and stored in each Gaussian point (iex,igx).

### 4.3.2 Evolution

In order to compute, after each time increment, the evolution of the internal stresses due to the misfits between the two distinct phases in single crystal superalloys, the subroutine **sub_int_evolution** located in Module **mod_wkcoup.f** is called.

The values of the eigenstrains $\boldsymbol{\epsilon}$ between phases are updated for each RVE subsection by adding up the newly computed eigenstrains to the values from previous time step. The phase fractions and the internal stresses are also updated for each subsection of the RVE. The evaluation of the internal stresses follows the linear relation based on the Eshelby inclusion approach, using the updated modules of MatrixA and updated eigenstrains [10]. In **mod_stress.f**, the plastic strain due to lattice mismatch are treated as eigenstrains $\boldsymbol{\epsilon}$ [12] according to the following Equations [9]:

$$\boldsymbol{\epsilon}_{\mathrm{x}} = \sum_{\alpha=1}^{12} \gamma^{\alpha}_{\mathrm{x110}}\mathbf{M}^{\alpha}_{<110>} + \sum_{\alpha=1}^{2} \gamma^{\alpha}_{\mathrm{xc}}\mathbf{M}^{\alpha}_{\mathrm{c}} \tag{4.21}$$

$$\boldsymbol{\epsilon}_{\mathrm{y}} = \sum_{\alpha=1}^{12} \gamma^{\alpha}_{\mathrm{y110}}\mathbf{M}^{\alpha}_{<110>} + \sum_{\alpha=3}^{4} \gamma^{\alpha}_{\mathrm{yc}}\mathbf{M}^{\alpha}_{\mathrm{c}} \tag{4.22}$$

$$\boldsymbol{\epsilon}_{\mathrm{z}} = \sum_{\alpha=1}^{12} \gamma^{\alpha}_{\mathrm{z110}}\mathbf{M}^{\alpha}_{<110>} + \sum_{\alpha=5}^{6} \gamma^{\alpha}_{\mathrm{zc}}\mathbf{M}^{\alpha}_{\mathrm{c}} \tag{4.23}$$

$$\boldsymbol{\epsilon}_{\mathrm{p}} = (1 - \sum_{\beta=1}^{12} f^{\beta})(\sum_{\alpha=1}^{12} \gamma^{\alpha}_{\mathrm{p112}}\mathbf{M}^{\alpha}_{<112>} + \sum_{\alpha=1}^{6} \gamma^{\alpha}_{\mathrm{pc}}\mathbf{M}^{\alpha}_{\mathrm{c}}) + \delta\mathbf{I} \tag{4.24}$$

Finally, the data associated with the internal stresses are stored in each material point (iex,igx).

### 4.3.3 Effect

If the additional eigendeformation due to Internal stresses is needed to be included as it is the case in Single Crystal Superalloys, the subroutine **sub_int_effect** is called to simply store the updated values of the internal stresses and phase fractions defined in each FEM grid point into the UMAT state variables after every time step.

# Chapter 5

# ABAQUS set-up and executive command

For a given model, in ABAQUS:

- In the property module, for the material definition, the number of solution-dependent state variables (Depvar) must be set according to the *statev* defined in **umat.f**, as shown in figure 5.1. According to the serial number of materials defined in UMAT, the material of interest (ialloy) is chosen as a number and filled in the first box of the User Material mechanical constants in figure 5.2. In the following three boxes, the initial Euler angles (eang00) are set.

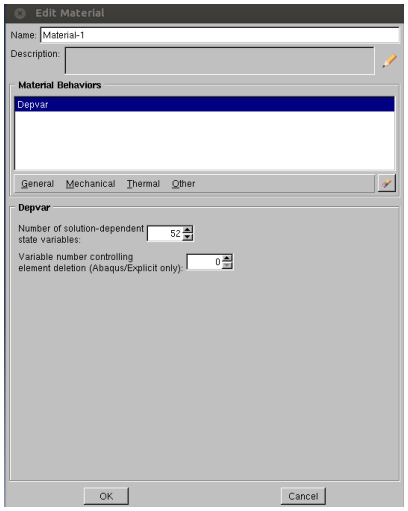


Figure 5.1: Set-up of solution-dependent state variables in ABAQUS.

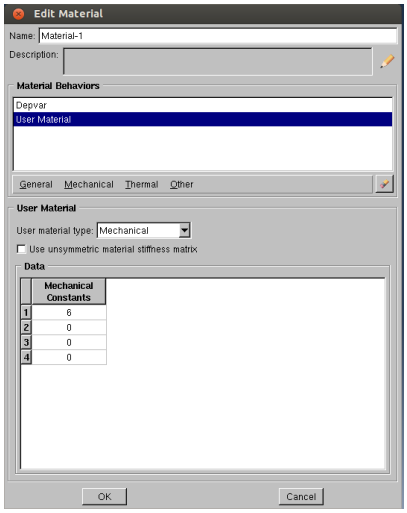


Figure 5.2: Set-up of User Material mechanical constants in ABAQUS.

- In the mesh module, the element type and element number must be consistent with that defined in the Module **mod_mesh_grad_size** in **mod_wkcoup.f**. (e.g., C3D8 is commonly used element type for UMAT.)
- The input file created from ABAQUS must be in the same directory as all the UMAT files.
- Run the calculation in the terminal by the command as: abq"version".exe (e.g., abq6111.exe) job=input_file_name (e.g., job-1) user=umat.f cups=cups_name int.
- An iterative solver must be set up to use multiple CPUs.

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